

● EASY

● ALGEBRA

Algebra

10 questions · ~15 min

18

Q1

ALGEBRA

$$s + 7r = 27$$

$$r = 3$$

What is the solution r, s to the given system of equations?

- A. $(6, 3)$
- B. $(3, 6)$
- C. $(3, 27)$
- D. $(27, 3)$

Vivian bought party hats and cupcakes for \$ 71. Each package of party hats cost \$ 3, and each cupcake cost \$ 1. If Vivian bought 10 packages of party hats, how many cupcakes did she buy?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

If $x = 40$, what is the value of $x + 6$?

A. 34

B. 40

C. 46

D. 64

Q4

GRID-IN

ALGEBRA

If $y = 5x + 10$, what is the value of y when $x = 8$?

STUDENT-PRODUCED RESPONSE

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.	/	.	.
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0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5

ALGEBRA

$$F = 2.50x + 7.00y$$

In the equation above, F represents the total amount of money, in dollars, a food truck charges for x drinks and y salads. The price, in dollars, of each drink is the same, and the price, in dollars, of each salad is the same. Which of the following is the best interpretation for the number 7.00 in this context?

- A. The price, in dollars, of one drink
- B. The price, in dollars, of one salad
- C. The number of drinks bought during the day
- D. The number of salads bought during the day

Q6

ALGEBRA

$$4x + 1 = 33$$

Which equation has the same solution as the given equation?

- A. $4x = 32$
- B. $4x = 5$
- C. $4x = 1$
- D. $4x = -32$

Q7

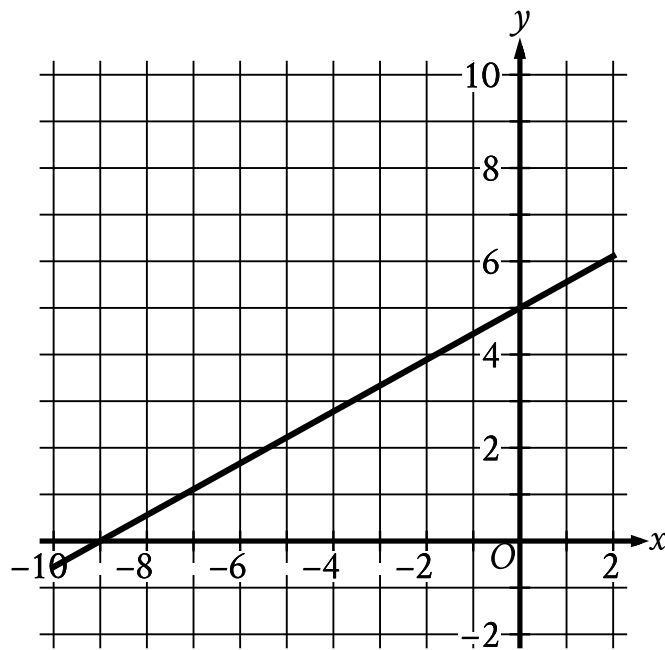
GRID-IN ALGEBRA

The function g is defined by $gx = 6x$. For what value of x is $g(x) = 54$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



- The line slants gradually up from left to right.
- The line passes through the following points:
 - (negative 9 comma 0)
 - (0 comma 5)

What is the y -intercept of the line graphed?

- A. -5,0
- B. 0,0
- C. 0,5
- D. 0,9

Q9

ALGEBRA

$$(p + 3) + 8 = 10$$

What value of p is the solution to the given equation?

- A. -1
- B. 5
- C. 15
- D. 21

Q10

ALGEBRA

On January 1, 2015, a city's minimum hourly wage was \$9.25. It will increase by \$0.50 on the first day of the year for the next 5 years. Which of the following functions best models the minimum hourly wage, in dollars, x years after January 1, 2015, where $x = 1, 2, 3, 4, 5$?

- A. $f(x) = 9.25 - 0.50x$
- B. $f(x) = 9.25x - 0.50$
- C. $f(x) = 9.25 + 0.50x$
- D. $f(x) = 9.25x + 0.50$